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Assignment - 4

Aim: Write a java program to showcase the use of File I/O exeution Exception Handling in java.

Objectives
- To study I/O stream classes
- Understand the different types of exceptions
- Exception handling using various exception classes.

Theory

1. Java I/O : input-output stream classes

Java I/O focuses on java.io package to process data through streams. Types:
- Byte Streams
- character streams
- File I/O

2. Exception handling mechanism

Try: identify & run main code block
catch: Handles exception
finally: executes essential codes irrespective of outcome
throw: used to signal or declare exception.

3. Types of exception classes

- checked
- unchecked
- standard
- user defined

Platform: Open source java programming tool like Eclipse Editor / Netbeans.

Conclusion: Thus, studied the exception handling concepts in java & implemented usage of I/O stream classes

FAQs

1. What is the difference between error & exception?

Soln: Error: indicates serious problems that a reasonable application should not try to catch, usually related to env.

Exception: indicates conditions that an application might want to catch & handle.
often caused by program logic.

2. Write a few examples for checked & unchecked exceptions?

Soln: Checked exceptions: - Occur at compile time.
ex:- IOException, SQLException

Unchecked exceptions: - Occur at run time
ex:- Arithmetic Expression, NullPointerException, Array Index Out Of Bound Exception.

3. What are 3 diff. keywords in exception handling in Java?

Ans: Java uses specific keywords:
- try
- catch
- finally
- throw
- throws

4. What is the diff. b/w throws & throw keywords in Java?

Soln: throw : An action keyword used within a method to actually "throw" a specific instance of an exception.

throws: A declarative keyword used in the method signature to warn caller.

=> throw is for object
=> throws is for class.

5. List down various byte & character stream classes in Java.

Ans: Byte Streams: File Input Stream, File Output Streams, Buffered Input Stream, Print Stream.

Character Streams: File Reader, File Writer, Buffered Reader, Buffered Writer.